

West Bengal Council of Higher Secondary Education

HIGHER SECONDARY EXAMINATION

Modern Computer Application (COMA)

Class XII – Theory

Full Marks : 70

Time : 3 Hours 15 Minutes

Special credit will be given for answers which are brief and to the point. Marks will be deducted for spelling mistakes, untidiness and bad handwriting.

Figures in the margin indicate full marks for the questions.

GROUP – A (Multiple Choice Type Questions)

1. Choose the correct option for each question : [21 × 1 = 21]

1. Artificial Intelligence is the science of making machines that can —

- | | |
|---------------------------------|---------------------------|
| (a) only store data | (c) only print |
| (b) simulate human intelligence | (d) only transmit signals |

2. The term “Artificial Intelligence” was first coined in the year —

- | | |
|----------|----------|
| (a) 1943 | (c) 1969 |
| (b) 1956 | (d) 1997 |

3. Who is regarded as the “Father of AI”?

- | | |
|---------------------|--------------------|
| (a) Charles Babbage | (c) Alan Turing |
| (b) John McCarthy | (d) Dennis Ritchie |

4. The test used to judge whether a machine can behave intelligently is the —

- | | |
|-----------------|------------------|
| (a) Litmus Test | (c) IQ Test |
| (b) Turing Test | (d) Boolean Test |

5. Learning from labelled data is called —

- | | |
|---------------------------|----------------------------|
| (a) Unsupervised learning | (c) Reinforcement learning |
| (b) Supervised learning | (d) Deep sleep learning |

6. Learning through rewards and penalties is —

- (a) Supervised (c) Reinforcement
(b) Unsupervised (d) Semi-supervised
7. A neural network with many hidden layers is the basis of —
(a) Data mining (c) OLAP
(b) Deep learning (d) DBMS
8. NLP stands for —
(a) Natural Language Processing (c) Network Layer Protocol
(b) Neural Logic Programming (d) Non-Linear Programming
9. Enabling a computer to “see” and interpret images is called —
(a) Computer Vision (c) Cloud Computing
(b) Computer Graphics (d) Data Warehousing
10. Detecting unusual credit-card transactions is an example of AI in —
(a) Education (c) Gaming
(b) Fraud detection (d) Weather
11. Which of the following is NOT a type of Machine Learning?
(a) Supervised (c) Reinforcement
(b) Unsupervised (d) Productive
12. AlphaGo or a chess-playing AI uses which type of learning?
(a) Supervised (c) Unsupervised
(b) Reinforcement (d) Batch
13. Clustering is an example of which type of learning?
(a) Supervised (c) Reinforcement
(b) Unsupervised (d) Labelled
14. A chatbot is mainly used in which field of AI?
(a) Agriculture (c) Mining
(b) Customer service (d) Weather
15. Deep learning is inspired by the structure of —
(a) the human brain (neural network) (c) the keyboard
(b) the hard disk (d) the printer
16. The main cause of bias in an AI system is —

- (a) a fast processor (c) a large monitor
(b) biased training data (d) more RAM
17. A self-driving car identifies roads and pedestrians using —
- (a) Computer Vision (c) OLAP
(b) DBMS (d) HTML
18. Algorithmic trading is an application of AI in which field?
- (a) Education (c) Entertainment
(b) Finance (d) Transport
19. Personalized learning and intelligent tutoring are applications of AI in —
- (a) Finance (c) Agriculture
(b) Education (d) Sports
20. Which of the following is an ethical concern of AI?
- (a) Bias and discrimination (c) Low cost
(b) Fast speed (d) Good graphics
21. Google Translate is an example of —
- (a) NLP (c) DBMS
(b) OLAP (d) DDL

GROUP – B (Short Answer Type Questions)

2. Answer the following questions :

$$[14 \times 1 = 14]$$

1. Define Artificial Intelligence.
2. State any two important milestones in the history of AI.
3. Write any two differences between human intelligence and artificial intelligence.
4. What is Machine Learning?
5. Distinguish between supervised and unsupervised learning.
6. What is reinforcement learning? Give one example.
7. What is a Neural Network?
8. What is Deep Learning?
9. What is Natural Language Processing (NLP)? Give one example.
10. What is Computer Vision?
11. Mention two applications of AI in finance.
12. What is a chatbot? Where is it used?

- 13.** What is meant by bias in an AI system?
- 14.** State two impacts of AI on employment.

GROUP – C (Broad / Descriptive Type Questions)**Answer any *five* of the following questions :****[5 × 7 = 35]**

1. Answer the following : **[4 + 3]**
 - (a) Define Artificial Intelligence and explain its scope.
 - (b) Differentiate AI from human intelligence with examples.
2. Answer the following : **[4 + 3]**
 - (a) Explain the three main types of Machine Learning.
 - (b) With a diagram, describe how a neural network works.
3. Answer the following : **[3 + 4]**
 - (a) What is Deep Learning?
 - (b) Explain NLP and Computer Vision with real-life examples.
4. Answer the following : **[4 + 3]**
 - (a) Describe any three applications of AI in finance.
 - (b) Explain how AI helps in education through personalized learning and intelligent tutoring.
5. Answer the following : **[3 + 4]**
 - (a) Discuss bias and fairness in AI systems.
 - (b) Explain the impact of AI on employment and social inequality.
6. Answer the following : **[4 + 3]**
 - (a) Give a historical overview of AI with key milestones.
 - (b) Explain the role of AI in customer service.
7. Answer the following : **[3 + 4]**
 - (a) Explain the ethical concerns of AI.
 - (b) Suggest measures to make AI systems fair and unbiased.

END OF QUESTION PAPER

*Prepared as per WBCHSE question pattern (Group A : 21, Group B : 14, Group C : 35
= 70 marks)*